AMOD-M motors and actuators

Jason Gallicchio and Claude, rev A, 2026-10-25 github.com/gallicchio/AMOD-and-RFMOD

FOOT FOOT FOOT FOOT

m2 Motor supply
5-16 V, centre +
or USB 5 V alone
VMOT = in - 0.4 V
not switched

m5 Servo
CMD 1-2 ms / 20 ms
header: S + -
I 1 V/A

m6 Solenoid
CMD high: coil on
jumper 1-2: diode
2-3: 24 V TVS
none: avalanche
VD = drain / 52

m1 Power
USB-C 5 V only
+5 V 1.5 A
-5 V 40 mA
+3.3 V 300 mA

m11 Load cell
1 red E+ 2 black E-
3 white S-
4 green S+
gain 201

m10 Optical
target 1 mm up
OUT 0.1-0.9 V
sensor top left

m9 Hall
9.75 mV/mT
+-85 mT
sensor top left

m7 Linear amp
load = 4 x IN
IN +-1 V
1 W in the chip:
8 ohm at 5-6 V
I 1 V/A both ways

github.com/gallicchio/
AMOD-and-RFMOD
handbook and generator

m4 Stepper
A: IN1 IO2, IN2 IO3
B: IN1 IO4, IN2 IO8
1 A per coil
XH: A+ A- B+ B-

m8 Buck stage
PWM 100-500 kHz
VO = duty x VMOT
load up to 1 A
I 1 V/A, VO /16

m3 DC motor
IN1 IO7, IN2 IO1
2 A limit
I 0.98 V/A
V = motor / 13
PH: M1 G A B 3V3 M2

AMOD-M: motors. Motor supply
5-16 V on m2; USB-C 5 V on m1.

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